

The major problem in the subject preference data is that we have the data for preference of cheeses, but the preference about dairy cheese alone are too sparse as they are, so we'll reduce dimensions and extract some meaningful features that better represent the signal.

```
library(psych)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tidyr)
library(ggplot2)
```

```
##
## Attaching package: 'ggplot2'

## The following objects are masked from 'package:psych':
##
##   %+%, alpha
```

```
library(factoextra)
```

```
## Welcome to factoextra!
## Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/practice
```

```
library(FactoMineR)
```

```
df <- read.csv("Subject_Data.csv")
head(df)
```

```
##   Unique_Panelist_ID Cheese_eater_type Frequency_of_dairy_cheese_consumption
## 1          442045             1                      5
## 2          442144             1                      4
## 3          443573             1                      6
## 4          444181             1                      6
## 5          445383             1                      7
## 6          445393             1                      7
##   Frequency_of_PB_cheese Frequency_PB Diet Age Experience_in_plant.based_cheese
## 1             1             1  1  2                      2
## 2             4             2  1  2                      1
## 3             6             2  1  2                      1
## 4             3             2  2  2                      1
## 5             5             2  1  3                      1
## 6             2             1  1  3                      1
##   Q11_1__Cheddar Q11_2__American Q11_3__Parmesan Q11_4__Gouda Q11_5__Mozzarella
## 1             1             1             1             0             1
## 2             1             1             1             0             1
## 3             1             1             1             1             1
## 4             1             1             1             1             1
## 5             1             1             1             1             1
```

```
## 6          1          0          1          1          1
##   Q11_6__Blue_Cheese Q11_7__Pepper_Jack Q11_8__Feta Q11_9__Swiss
## 1          1          0          1          1          1
## 2          0          1          1          1          0
## 3          1          1          0          0          1
## 4          0          1          0          0          0
## 5          1          1          1          1          1
## 6          1          1          1          1          1
##   Q11_10__Cream_cheese
## 1          1
## 2          1
## 3          1
## 4          1
## 5          1
## 6          1
```

```
cheese_vars <- colnames(df)[9:18]

X <- df %>%
  dplyr::select(all_of(cheese_vars))

#Sanity Check: Ensuring 0-1 coding
X <- X %>%
  mutate(across(everything(), ~ as.numeric(as.character(.))))

# Optional sanity checks
stopifnot(all(sapply(X, function(z) all(z %in% c(0, 1, NA)))))
```

Since these variables have a dichotomous structure, we will use tetrachoric correlation, it assumes there is reason you choose 0 or 1, i.e. there is a continuous variable which represents your tendency to like a certain cheese, so a herarchical model, which is giving you 0 or 1 now, we are looking at the correlation between these hidden tendency variables. Given 0 your tendency comes from distribution A and given 1, tendency comes from distribution Y. We will find correlation between X and Y.

Likelihood of liking each dairy cheese:

```
like_rates <- colMeans(X, na.rm = TRUE)
print(round(like_rates, 3))
```

```
##      Q11_1__Cheddar      Q11_2__American      Q11_3__Parmesan
##      0.919            0.541            0.748
##      Q11_4__Gouda      Q11_5__Mozzarella      Q11_6__Blue_Cheese
##      0.437            0.837            0.319
##      Q11_7__Pepper_Jack      Q11_8__Feta      Q11_9__Swiss
##      0.474            0.519            0.519
##      Q11_10__Cream_cheese
##      0.674
```

Tetrachoric Correlation Matrix:

```
tet_out <- psych::tetrachoric(as.matrix(X))
```

```
## For i = 4 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!
## For i = 6 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!
## For i = 7 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!
## For i = 8 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!
```

```
## Warning in cor.smooth(mat): Matrix was not positive definite, smoothing was
## done
```

```
R_tet <- tet_out$rho
print(round(R_tet, 2))
```

```
##          Q11_1__Cheddar Q11_2__American Q11_3__Parmesan
## Q11_1__Cheddar          1.00          0.36          0.53
## Q11_2__American          0.36          1.00          0.36
## Q11_3__Parmesan          0.53          0.36          1.00
## Q11_4__Gouda            0.65          0.55          0.71
## Q11_5__Mozzarella        0.04          0.36          0.64
## Q11_6__Blue_Cheese       0.54          0.35          0.74
## Q11_7__Pepper_Jack       0.68          0.34          0.63
## Q11_8__Feta              0.71          0.37          0.79
## Q11_9__Swiss             0.62          0.42          0.49
## Q11_10__Cream_cheese     0.30          0.53          0.74
##          Q11_4__Gouda Q11_5__Mozzarella Q11_6__Blue_Cheese
## Q11_1__Cheddar          0.65          0.04          0.54
## Q11_2__American          0.55          0.36          0.35
## Q11_3__Parmesan          0.71          0.64          0.74
## Q11_4__Gouda            1.00          0.53          0.82
## Q11_5__Mozzarella        0.53          1.00          0.61
## Q11_6__Blue_Cheese       0.82          0.61          1.00
## Q11_7__Pepper_Jack       0.61          0.49          0.53
## Q11_8__Feta              0.77          0.48          0.63
## Q11_9__Swiss             0.63          0.33          0.63
## Q11_10__Cream_cheese     0.67          0.74          0.67
##          Q11_7__Pepper_Jack Q11_8__Feta Q11_9__Swiss
## Q11_1__Cheddar          0.68          0.71          0.62
## Q11_2__American          0.34          0.37          0.42
## Q11_3__Parmesan          0.63          0.79          0.49
## Q11_4__Gouda            0.61          0.77          0.63
## Q11_5__Mozzarella        0.49          0.48          0.33
## Q11_6__Blue_Cheese       0.53          0.63          0.63
## Q11_7__Pepper_Jack       1.00          0.49          0.48
## Q11_8__Feta              0.49          1.00          0.52
## Q11_9__Swiss             0.48          0.52          1.00
## Q11_10__Cream_cheese     0.45          0.58          0.62
##          Q11_10__Cream_cheese
## Q11_1__Cheddar          0.30
## Q11_2__American          0.53
## Q11_3__Parmesan          0.74
## Q11_4__Gouda            0.67
## Q11_5__Mozzarella        0.74
## Q11_6__Blue_Cheese       0.67
## Q11_7__Pepper_Jack       0.45
## Q11_8__Feta              0.58
## Q11_9__Swiss             0.62
## Q11_10__Cream_cheese     1.00
```

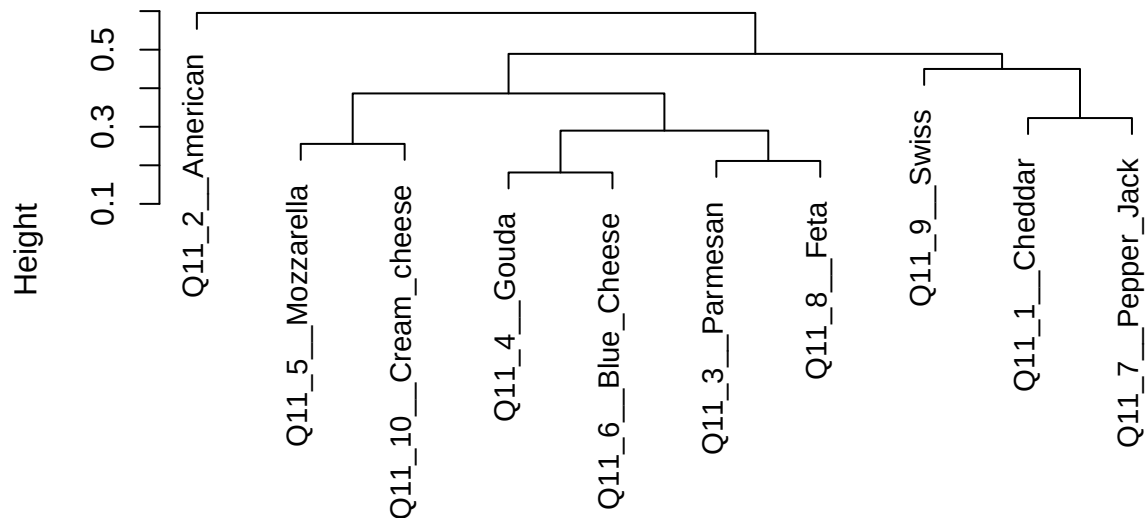
Once we have these we will perform hierarchical clustering to see which cheeses are similar to each other, using a dendrogram. Distance metric for similarity is 1-r, it converts strong positive correlation into small distance, so cheeses with highly similar liking patterns are placed close together and are likely to merge early in the dendrogram. if Cheddar and Gouda have a high tetrachoric correlation, then 1-r is small, so the

algorithm treats them as neighbors. If Blue Cheese and American have weak or negative association, then their distance is larger, so they stay apart until much later.

```
D <- as.dist(1 - R_tet)
hc <- hclust(D, method = "average")

plot(hc, main = "Cheese clustering from tetrachoric correlations",
      xlab = "", sub = "", cex = 0.9)
```

Cheese clustering from tetrachoric correlations



Things that are in the same child node are very similar to each other (theoretically)

We will go with 4 clusters based on visual checks of the dendrogram.

```
cluster_4 <- cutree(hc, k = 4)
print(cluster_4)
```

```
##      Q11_1__Cheddar      Q11_2__American      Q11_3__Parmesan
##              1              2              3
##      Q11_4__Gouda      Q11_5__Mozzarella      Q11_6__Blue_Cheese
##              3              3              3
##      Q11_7__Pepper_Jack      Q11_8__Feta      Q11_9__Swiss
##              1              3              4
##      Q11_10__Cream_cheese
##              3
```

Factor Analysis uncovers hidden variables that represent real relationships. What broader taste traits explain the whole response pattern?

```
fa2 <- psych::fa(R_tet, nfactors = 2, n.obs = nrow(X), fm = "minres", rotate = "oblimin")
```

```

## Loading required namespace: GPArotation

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully
fa3 <- psych::fa(R_tet, nfactors = 3, n.obs = nrow(X), fm = "minres", rotate = "oblimin")

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully
fa4 <- psych::fa(R_tet, nfactors = 4, n.obs = nrow(X), fm = "minres", rotate = "oblimin")

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully
print(fa2$loadings, cutoff = 0.25)

##
## Loadings:
##
##          MR1      MR2
## Q11_1__Cheddar          1.035
## Q11_2__American      0.417
## Q11_3__Parmesan      0.720  0.264
## Q11_4__Gouda         0.613  0.458
## Q11_5__Mozzarella     0.966 -0.293
## Q11_6__Blue_Cheese    0.672  0.299
## Q11_7__Pepper_Jack    0.404  0.436
## Q11_8__Feta           0.490  0.484
## Q11_9__Swiss          0.408  0.440
## Q11_10__Cream_cheese  0.898
##
##          MR1      MR2
## SS loadings      3.828  2.180
## Proportion Var  0.383  0.218
## Cumulative Var  0.383  0.601
print(fa3$loadings, cutoff = 0.25)

##
## Loadings:
##
##          MR3      MR1      MR2
## Q11_1__Cheddar          0.939
## Q11_2__American      0.610
## Q11_3__Parmesan          0.890
## Q11_4__Gouda         0.495  0.300  0.329
## Q11_5__Mozzarella     0.362  0.633 -0.380
## Q11_6__Blue_Cheese    0.399  0.419
## Q11_7__Pepper_Jack    0.410  0.349
## Q11_8__Feta           0.577  0.389
## Q11_9__Swiss          0.713      0.343
## Q11_10__Cream_cheese  0.783
##
##          MR3      MR1      MR2
## SS loadings      2.064  2.058  1.622
## Proportion Var  0.206  0.206  0.162
## Cumulative Var  0.206  0.412  0.575

```

```
print(fa4$loadings, cutoff = 0.25)
```

```
##
## Loadings:
##           MR1    MR2    MR4    MR3
## Q11_1__Cheddar           0.921
## Q11_2__American           1.015
## Q11_3__Parmesan    0.793  0.328
## Q11_4__Gouda       0.475  0.358
## Q11_5__Mozzarella  0.907 -0.251
## Q11_6__Blue_Cheese 0.589           0.291
## Q11_7__Pepper_Jack 0.386  0.412
## Q11_8__Feta       0.514  0.506
## Q11_9__Swiss           0.959
## Q11_10__Cream_cheese 0.657           0.307
```

```
##
##           MR1    MR2    MR4    MR3
## SS loadings    2.872 1.641 1.161 1.120
## Proportion Var 0.287 0.164 0.116 0.112
## Cumulative Var 0.287 0.451 0.567 0.679
```

```
fa.parallel(x = as.matrix(X), cor = "tet", fa = "fa", n.iter = 50)
```

```
## For i = 4 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!
## For i = 6 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!
## For i = 7 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!
## For i = 8 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!
## Warning in cor.smooth(mat): Matrix was not positive definite, smoothing was
## done
## For i = 10 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!
## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully
## For i = 3 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!
## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully
## For i = 6 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!
## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully
## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully
## For i = 3 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!
## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully
## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully
## For i = 5 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!
```

```
## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully

## For i = 3 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully

## For i = 10 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully

## For i = 5 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully

## For i = 5 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully

## For i = 8 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully

## For i = 5 j = 1 A cell entry of 0 was replaced with correct = 0.5. Check your data!

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully

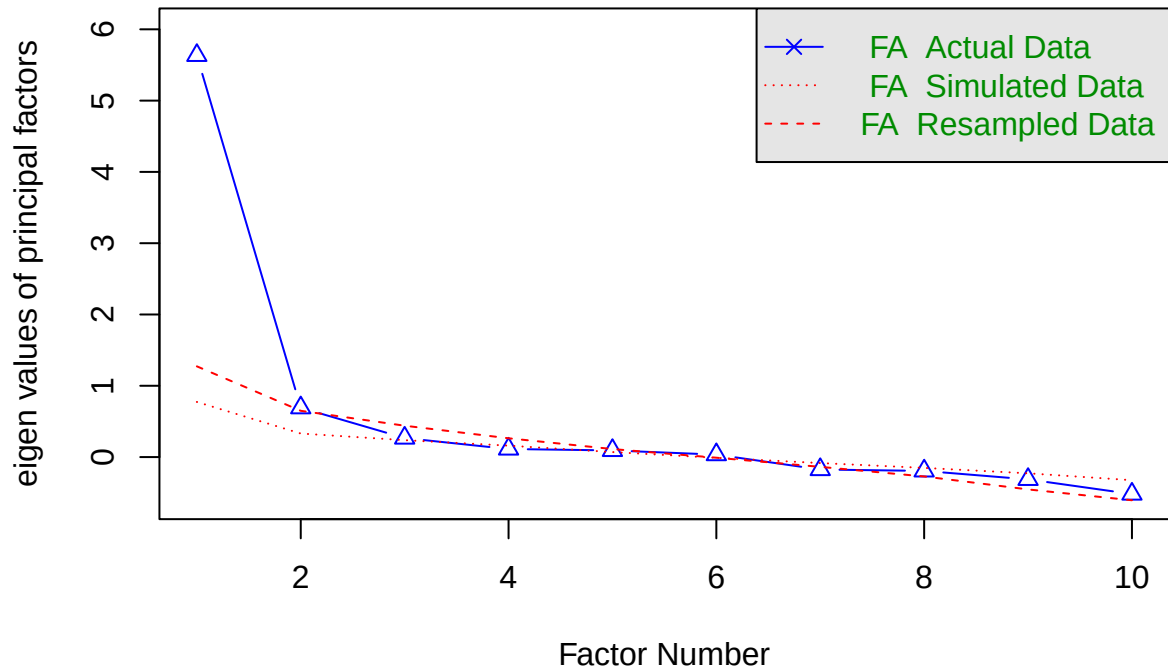
## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully

## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully
```

Parallel Analysis Scree Plots



Parallel analysis suggests that the number of factors = 2 and the number of components = NA

Parallel analysis indicates use of 2 latent factors, eigen values from 3rd factor on are somewhat higher for the resampled and simulated data than our actual data, so they will probably not add much information.

We will use 2 latent factors, the 2 latent factors also explain about 60% of the variability.

```
fa2 <- fa(R_tet, nfactors = 2, n.obs = nrow(X), fm = "minres", rotate = "oblimin")
```

Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An

ultra-Heywood case was detected. Examine the results carefully

```
print(fa2$loadings, cutoff = 0.30)
```

##

Loadings:

	MR1	MR2
## Q11_1__Cheddar		1.035
## Q11_2__American	0.417	
## Q11_3__Parmesan	0.720	
## Q11_4__Gouda	0.613	0.458
## Q11_5__Mozzarella	0.966	
## Q11_6__Blue_Cheese	0.672	
## Q11_7__Pepper_Jack	0.404	0.436
## Q11_8__Feta	0.490	0.484
## Q11_9__Swiss	0.408	0.440
## Q11_10__Cream_cheese	0.898	

##

	MR1	MR2
--	-----	-----


```
## SS loadings    3.828 2.180
## Proportion Var 0.383 0.218
## Cumulative Var 0.383 0.601
```

```
print(fa2$Phi)
```

```
##           MR1           MR2
## MR1 1.0000000 0.3999223
## MR2 0.3999223 1.0000000
```

Mozzarella and cream cheese load very strongly on the first latent factor, while Parmesan, Gouda and Blue cheese also show moderately strong loadings on the first latent factor. Cheddar loads very strongly on the second latent factor, while Gouda, Pepper Jack, Feta and Swiss cross load moderately on both latent factors. It is difficult to see a real structure here due to the cross loadings so we'll use the latent factors as covariates and make inferences on them.

My opinion: The latent factors don't correspond to individual distinctive types of cheeses as I had hoped. Factor 1 may reflect broad openness to creamy and distinctive cheeses (blue cheese here does seem a little weird) and factor 2 may correspond to firmer, savory, cheeses which people may use on a daily basis. Factor 2 is kind of dominated by cheddar, this may be due to the fact that cheddar is universally liked and 91% of subjects like Cheddar Cheese, so this may not a

When making inferences, remember the dependence of individual cheeses of latent factors: MR1 : Mozzarella, Cream cheese, parmesan, blue cheese, Cream Cheese MR2: Cheddar, Gouda, Swiss, Pepper Jack, Feta

Gouda, Swiss, Pepper Jack and Feta cross load so it's difficult to say, based on feature importance scores of MR's, whether they are the reason. They are mixed items not pure indicators of either latent factor

PS: A 3 factor solution was considered too, but heavy correlation was observed between the factors and we can't orthogonally rotate these since I do expect their to be correlation in cheese types, also cheddar single handedly dominated an entire latent factor, we somewhat see that in 2 factors too, but it was much more severe. Adding the latent factors to our subject level data:

```
scores <- factor.scores(
  x = X,
  f = fa2,
  method = "tenBerge"
)$scores

df$Cheese_MR1 <- scores[,1]
df$Cheese_MR2 <- scores[,2]

head(df)
```

```
## Unique_Panelist_ID Cheese_eater_type Frequency_of_dairy_cheese_consumption
## 1 442045 1 5
## 2 442144 1 4
## 3 443573 1 6
## 4 444181 1 6
## 5 445383 1 7
## 6 445393 1 7
## Frequency_of_PB_cheese Frequency_PB Diet Age Experience_in_plant.based_cheese
## 1 1 1 1 2 2
## 2 4 2 1 2 1
## 3 6 2 1 2 1
## 4 3 2 2 2 1
## 5 5 2 1 3 1
## 6 2 1 1 3 1
```

```

##    Q11_1__Cheddar Q11_2__American Q11_3__Parmesan Q11_4__Gouda Q11_5__Mozzarella
## 1          1          1          1          0          1
## 2          1          1          1          0          1
## 3          1          1          1          1          1
## 4          1          1          1          1          1
## 5          1          1          1          1          1
## 6          1          0          1          1          1
##    Q11_6__Blue_Cheese Q11_7__Pepper_Jack Q11_8__Feta Q11_9__Swiss
## 1          1          0          1          1
## 2          0          1          1          0
## 3          1          1          0          1
## 4          0          1          0          0
## 5          1          1          1          1
## 6          1          1          1          1
##    Q11_10__Cream_cheese Cheese_MR1 Cheese_MR2
## 1          1 0.8498086 0.25009262
## 2          1 0.3073561 0.03957266
## 3          1 1.0929630 0.57291983
## 4          1 0.4949372 0.10651435
## 5          1 1.1563717 0.86691478
## 6          1 1.0205179 0.80781769

```

So going forward followings are the variables we should include for subject level data: 1. Unique Panelist ID
2. Frequency_of_PB_Cheese 3. Age 4. Cheese_MR1 5. Cheese_MR2

```
write.csv(df, "Subject_Level_Dairy_Cheese_latent_factors.csv")
```